

chunks, which can be integrated together to create a real world application.

Please note that the usage for basic Arduino™ specific programming syntax is explained at the end of the chapter under the section ‘Syntax for writing Arduino™ sketches’.

Let’s start our journey into the world of hardware programming with the “Hello World!” counterpart of the electronics world called “LED Blink”.

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As Arduino™ was originally aimed at designers, the Arduino™ programs are called ‘Sketches’.

LED blink

Every Arduino™ board has an LED connected to Pin 13. We’ll use this LED to write the first code. The best thing about this piece of code is that it doesn’t need any extra hardware to be attached to the Arduino™ board. Since a display is optional in the electronics world, the LED is commonly used as a debugging tool to check if things are working as you want them to work.

There’s no direct way to keep an LED blinking. To achieve this, we need to trigger the following algorithm:

1. Turn on the LED
2. Wait for some time
3. Then turn it off
4. Wait for some time
5. Again repeat Step 1

DECIPHERING THE LED BLINK CODE

<code>int led = 13;</code>	Set pin number where LED is connected using the label ‘led’
<code>void setup () { pinMode (led, OUTPUT); }</code>	Set the pin 13 where LED is connected as output pin
<code>void loop () {</code>	Start the loop to blink LED
<code> digitalWrite (led, HIGH);</code>	Turn ON the LED by writing logic ‘HIGH’ to LED pin
<code> delay (1000);</code>	Wait for 1000 milliseconds (i.e. 1 second)
<code> digitalWrite (led, LOW);</code>	Turn OFF the LED by writing logic ‘HIGH’ to LED pin
<code> delay (1000); }</code>	Wait for 1000 milliseconds (i.e. 1 second). Proceed to first statement of loop function i.e. digitalWrite (led, HIGH);